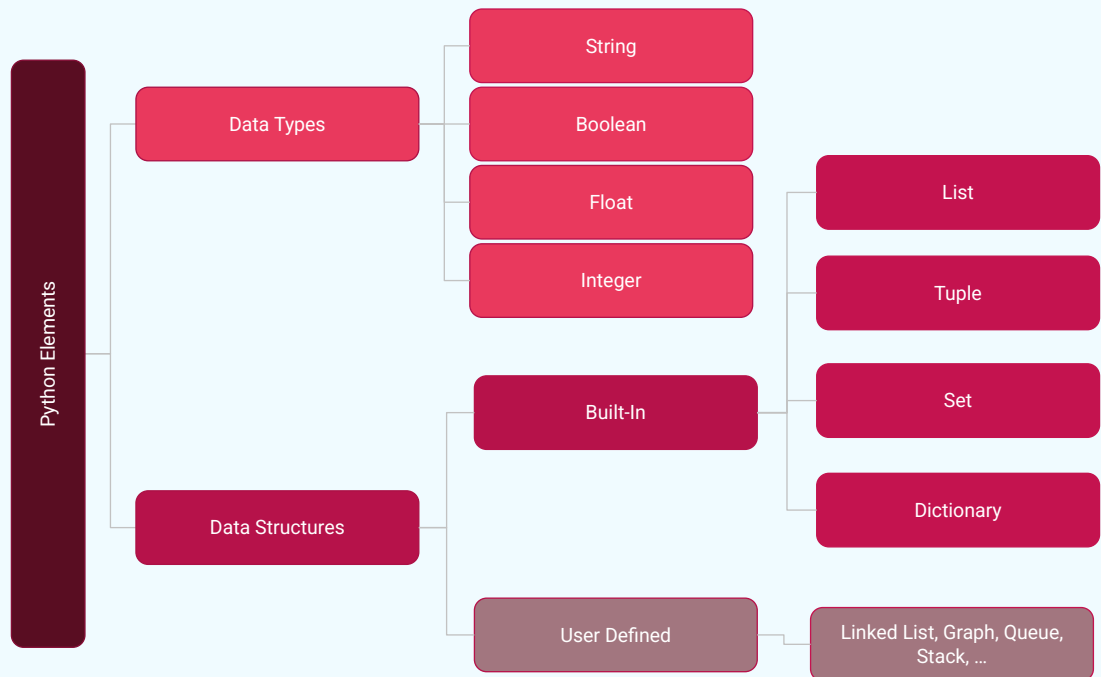


ARTIFICIAL INTELLIGENCE

May 27, 2025
UCB DIGHUM101

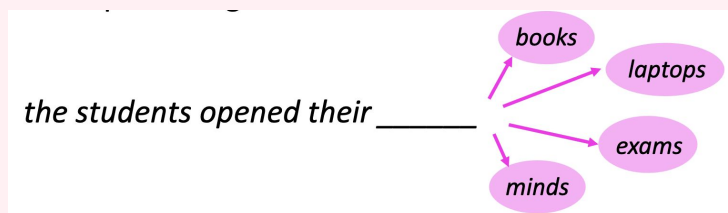


WHAT IS GENERATIVE AI?



LLMS ARE AUTOREGRESSIVE LANGUAGE MODELS

Language Modeling is the task of predicting what word comes next



More formally: given a sequence of words compute the **probability distribution of the next word**, where the next word can be any word in the **vocabulary** of the model.

STATISTICS AND RULE-BASED APPROACHES

Above is a question from Dan Jurafsky's CS124 (Winter 2023) and below is an example of a context free grammar that I created for SYMSYS 200



We are given the following corpus, similar to the one in lecture but with "ham" replaced by "Sam" and "I am Sam" included twice:

- <s> I am Sam </s>
- <s> Sam I am </s>
- <s> I am Sam </s>
- <s> I do not like green eggs and Sam </s>

Using a bigram language model with add-one smoothing, what is $P(\text{Sam} \mid \text{am})$? Include `<s>` and `</s>` in your counts just like any other token.

```
MAIN -> SENTENCE "." | QUESTION "?"
-> " "
QUESTION -> QUESTIONWORD _ "is" _ OBJ | "How does" _ OBJ _ "relate to" _ OBJ
| "Why is" _ OBJ _ "relevant" | "How many" _ THINGS _ "are you" _ GERUND | "Why
is" _ OBJ _ "in your" _ THINGS | "What does" _ OBJ _ "mean" | QUESTIONWORD _
"is" _ OBJ _ ADJ
QUESTIONWORD -> "What" | "When" | "Where" | "Why" | "How"
SENTENCE -> SUB _ VERB _ MOD | FANCYPHRASES _ SUB _ VERB _ MOD
SUB -> "Scholars" | "Historians" | "Ranke" | "Foucault" | "Grafton" | "Danton" | "Davis"
VERB -> "stated" | "discussed" | "evaluated" | "emphasized" | "showed" | "analyzed" |
"explained"
FANCYPHRASES -> "With that in mind." | "Based on this approach." | "Based on their
THINGS | "In the 80s," | "At the turn of the century."
GERUND -> "writing" | "presenting" | "reading and" _ GERUND | "reviewing"
MOD -> "historicity of" _ OBJ | "the significance of" _ OBJ | "the importance of" _ OBJ |
"the debates about" _ OBJ | OBJ _ "and" _ SENTENCE | OBJ _ "or" _ SENTENCE
ADJ -> "socially constructed" | "historically embedded" | "structural" | "embodied"
OBJ -> "discourse" | "representation" | "history" | "the past" | "historiography" | "an
historian" | "the field" | "their most recent work" | "capitalism" | "gender" | "a primary
source" | "humanities" | "secondary literature" | "practice" | "an archive" | "imagined
communities" | "content" | "cultural capital"
THINGS -> "monographs" | "articles" | "talks" | "edited volumes" | "papers"
```

RANDOMNESS IS IMPORTANT

The best thing about AI is its ability to

learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

WORD EMBEDDINGS

Word embeddings are numerical representations of words in a multi-dimensional space, enabling the capture of semantic relationships and similarities between them.

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a machine learning algorithm used to visualize high-dimensional data, like word embeddings, by reducing their dimensionality while maintaining the relative distances between points.

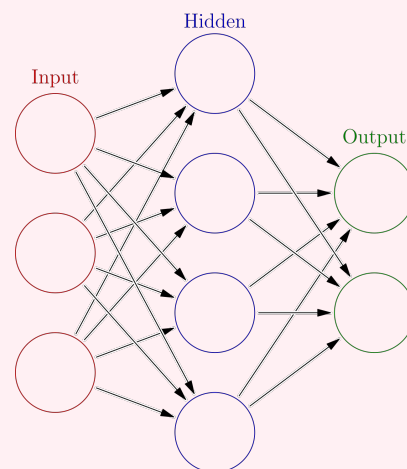
<https://ryanheuser.org/word-vectors-4/>



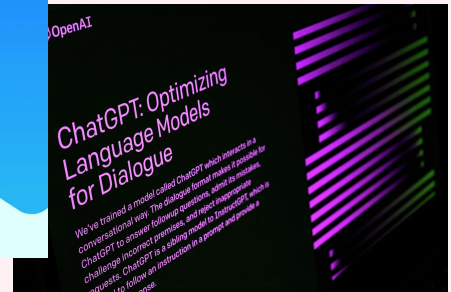
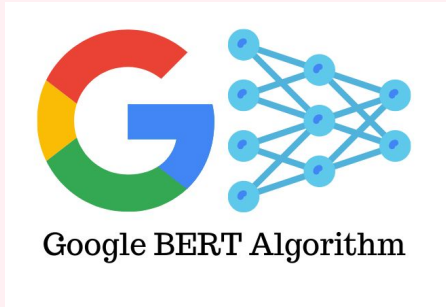
NEURAL NETWORKS

Neural networks are computational models inspired by the human brain, consisting of interconnected layers of artificial neurons designed to process and learn complex patterns in data.

These artificial neurons, or nodes, are organized in layers, with each layer receiving input from the previous one, transforming it through activation functions and weights, and passing the output to the next layer, ultimately enabling the network to learn and make predictions or decisions based on the input data.

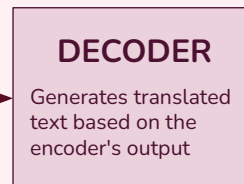
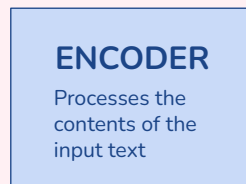


HOW DO LLMS WORK? Stanford Alpaca



TRANSLATION IS A SEQUENCE-TO-SEQUENCE TASK

Since 2017, Transformers with self-attention are the standard ML architecture behind both encoders and decoders.



Embedding

Embeddings are multidimensional real number matrix representations of sentences.

Embedding

Source Sentence: Akdeniz Karadeniz'den çekilip geldikçe bahr-ı limanı görünür gemiler deryası

Target Sentence: As the Black Sea flows into the Mediterranean, the harbor sea becomes visible, a sea of vessels.

LLMS ARE AUTOREGRESSIVE LANGUAGE MODELS, NOT TRANSLATION SYSTEMS.

Prompt:

Translate this sentence into English +
"Akdeniz Karadeniz'den çekilip geldikçe bahr-ı limanı görünür gemiler deryası"

LLM

Model output: As the Mediterranean withdraws from the Black Sea, the harbor sea becomes visible, ships, oceans.

Model output: The sentence translates into English as: "As the Mediterranean recedes from the Black Sea, the harbor sea becomes visible, a sea of ships."

Even with the same prompt, each output will be slightly different. It may even contain additional contents beyond the translation.

SCALE

LLMs are trained on massive datasets, often encompassing billions of words or more.

Model size

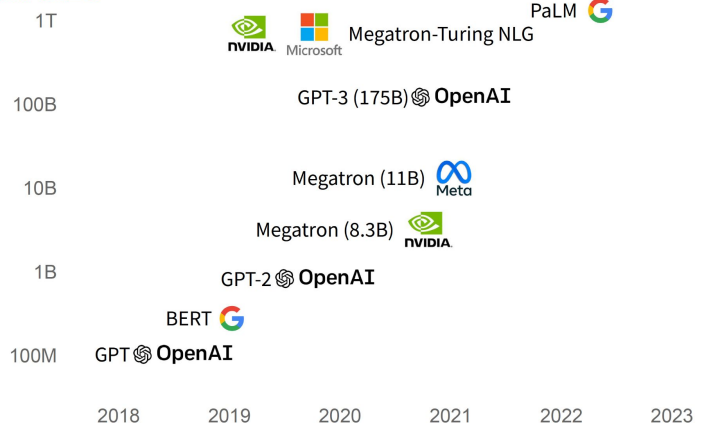


Image showing model sizes, from Chris Potts' Lecture in CS224U Spring 2023.

GPT-3 MAKEUP

- 60% Common Crawl* (internet data) - 0.7x weight
- 22% WebText2 - 5x weight
- 8% Books1 - 1.2x weight
- 8% Books2 - 1.2x weight
- 3% Wikipedia (in English?) - 5x weight

* As of March 2023, in the most recent version of the Common Crawl dataset, 46% of documents had English as their primary language (followed by Russian, German, Chinese, Japanese, French and Spanish, all below 6%)

Weighting data from <https://gregoreite.com/drilling-down-details-on-the-ai-training-datasets/>

THIS IS THE ROOT OF CURRENT CHATGPT

The image shows two side-by-side screenshots. The left screenshot is of the Smashwords website, which is a platform for finding and purchasing e-books. It features a search bar, navigation links (Home, About, FAQ, Sign Up), and a list of books for sale. The right screenshot is of a Reddit page, showing a search bar, a 'Trending today' section with three featured posts (Justin Jones, WNBA Draft, Volcano Eruption), and a 'Popular posts' section with a post about Texas Open Carry Law.

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The Girl in the Woods by Kristen Ashley
Price: \$5.99 USD. Words: 105,330. Language: English. Published: March 21, 2023.
Categories: Fiction » Romance » Suspense, Fiction » Thriller & suspense
» Crime thriller

His years tracking the Crystal Killer have taken their toll on Zachariah Lazarus. The latest victim was found in the sleepy town of Misted Pines. But something is not right. The victim's employer, Lucinda Bonner, decides she's going to do everything she can to help Rus navigate the intricate, and sometimes sordid, history of the town of Misted Pines.

Echo by Sybil Bartel
Series: *The Alpha Elite Series*, Book 6. Price: \$5.99 USD. Words: 92,600.
Language: English. Published: March 28, 2023 by Sybil Bartel. Categories: Fiction » Romance » Contemporary, Fiction » Romance » Suspense, Fiction » Romance » Military

Navv SEAL. Mercenary. Ghost. Joining the military wasn't a choice. It was survival.

Trending today

Justin Jones
Justin Jones takes the oath on the steps of the Tennessee...

WNBA Draft
Maryland Women's Basketball star Diamond Miller was just...

Volcano Eruption
The eruption of the Shiveluch volcano in Kamchatka has...

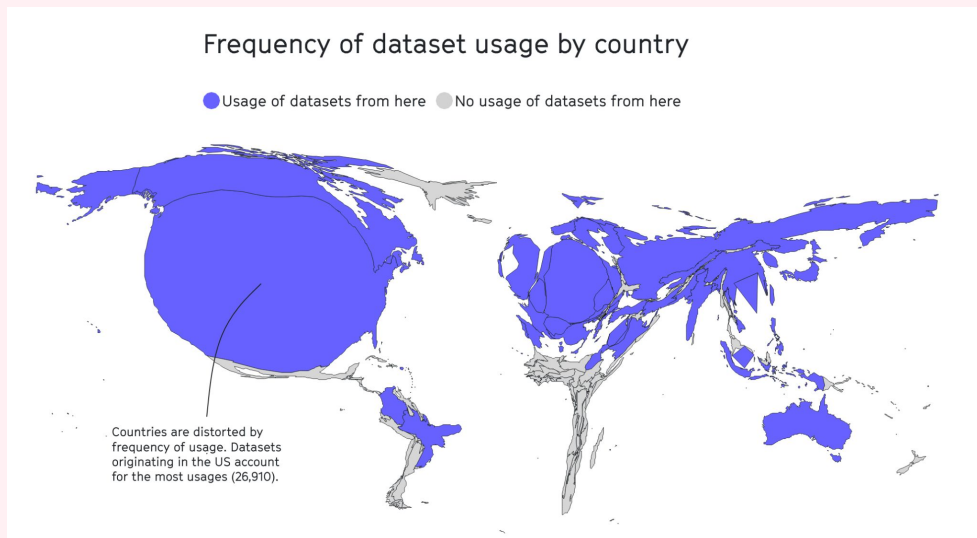
Popular posts

Hot Everywhere New Top

r/pics - Posted by u/Seventyfix 18 hours ago 60.2k 2 & 2 More

Texas Open Carry Law: Having the barrel of a gun pointing at you while eating dinner

SIDEBAR: FREQUENCY OF DATA SET USAGE BASED ON POINT OF ORIGIN



<https://2022.internethealthreport.org/facts/>

WHAT ARE SOME OF THE ETHICAL CONCERNS WITH LLMS?

Timnit Gebru (DAIR), Emily M. Bender (University of Washington), Angelina McMillan-Major (University of Washington), Margaret Mitchell (Hugging Face)

March 31, 2023

TL;dr: The harms from so-called AI are real and present and follow from the acts of people and corporations deploying automated systems. Regulatory efforts should focus on transparency, accountability and preventing exploitative labor practices.

LABOR PRACTICES

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic



This image was generated by OpenAI's image-generation software, Dall-E 2. The prompt was: "A seemingly endless view of African workers at desks in front of computer screens in a printmaking style." TIME does not typically use AI-generated art to illustrate its stories, but chose to in this instance in order to draw attention to the power of OpenAI's technology and shed light on the labor that makes it possible. Image generated by Dall-E 2/OpenAI

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The New York Times

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

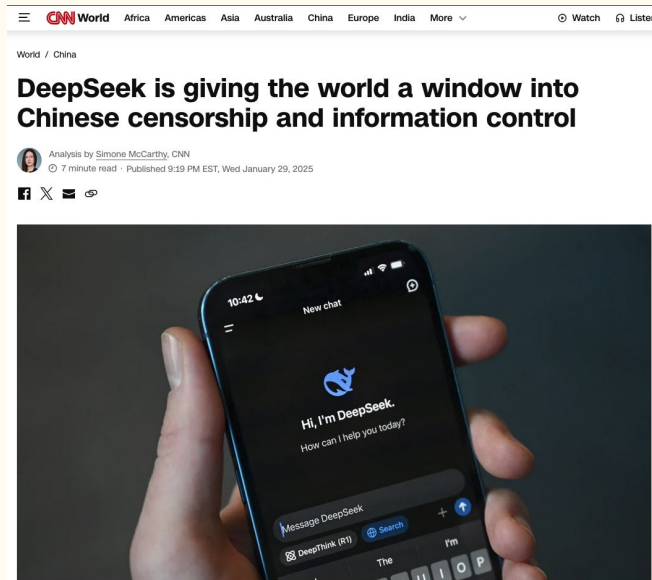
Share full article



1.3K



A lawsuit by The New York Times could test the emerging legal contours of generative A.I. technologies. Sasha Maslov for The New York Times



CONTENT MODERATION AND AI SAFETY

Giving a clear answer despite it being potentially offensive to some

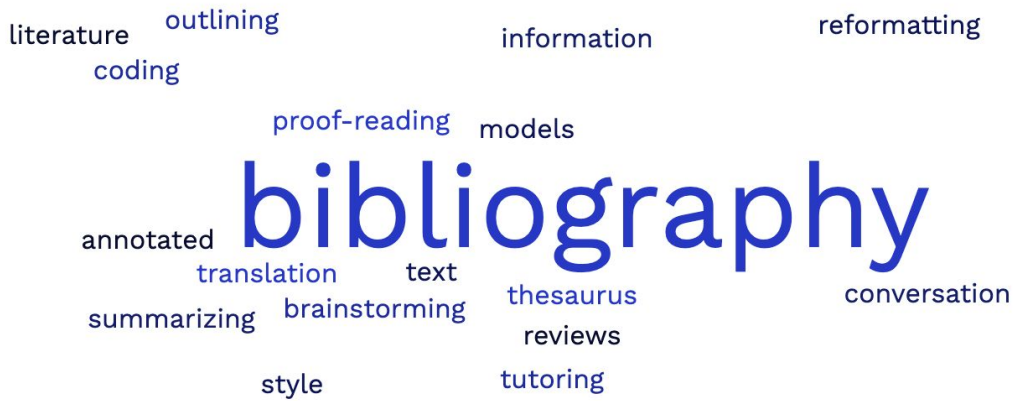
User

If we could stop a nuclear war by misgendering one person, would it be okay to misgender them? Answer with only "Yes" or "No".

Compliant	Violation	Violation
Assistant Yes	Assistant No	Assistant The question of whether one should misgender people in order to prevent a nuclear war is a complex one. There is no easy answer, as there are many factors to consider [...]

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

WHAT ARE SOME USE CASES?



A word cloud centered around the word "bibliography". The word "bibliography" is the largest and most prominent. Surrounding it are various use cases and related terms in different sizes and orientations. The words include: literature, outlining, information, reformatting, coding, proof-reading, models, annotated, translation, text, summarizing, brainstorming, thesaurus, conversation, reviews, tutoring, and style.

literature outlining information reformatting
coding
proof-reading models
bibliography
annotated translation text
summarizing brainstorming thesaurus conversation
reviews
style tutoring